

CS & IT ENGINEERING



Algorithms

Greedy Algorithms

Lecture No.- 01



By- Aditya sir

Recap of Previous Lecture



Topic

Topic

Divide & Conquer

Topics to be Covered



Topic

Topic

Topic

Greedy Algo



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

TTTe

[MCQ]

PYQ

#Q. For Constant $a \geq 1$ and $b > 1$, consider the following recurrence defined on the non-negative integers:

$$T(n) = a \cdot T\left(\frac{n}{b}\right) + f(n)$$

Which one of the following options is correct about the recurrence $T(n)$?

A

If $f(n)$ is $\theta(n^{\log_b(a)})$ then $T(n)$ is $\theta(n^{\log_b(a)})$

B

If $f(n)$ is $\theta(n^{\log_b(a)-e})$ for some $e > 0$, then $T(n)$ is $\theta(n^{\log_b(a)})$

C

If $f(n)$ is $\frac{n}{\log_2(n)}$ then $T(n)$ is $\theta(n \log_2(n))$

D

If $f(n)$ is $n \log_2(n)$, then $T(n)$ is $\theta(n \log_2(n))$

Ans: B

[MCQ]

#Q. When $n = 2^{2k}$ for some $k \geq 0$, then recurrence relation

$$T(n) = \sqrt{2}T(n/2) + \sqrt{n}, \quad T(1) = 1$$

Evaluates to :

A ~~$O(\log_2 \log_2 n)$~~

B ~~$O(\log_b \log_a n)$~~

C ~~$O(\log_a \log_b n)$~~

D ~~$O(\log_{ab} n)$~~

$$T(n) = \sqrt{2} T(n/2) + \sqrt{n} \quad \text{--- ①}$$

$$\begin{aligned} a &> 1 \\ b &> 1 \\ F(n) &= +\infty \end{aligned}$$

$$a = \sqrt{2}$$

$$b = 2$$

$$F(n) = \sqrt{n}$$

$$\log_b a = \log_2(\sqrt{2})$$

$$= \log_2(2^{1/2})$$

$$= \boxed{\frac{1}{2}}$$

Case 1: $f(n) = O(n^{\log_2 9 - \epsilon})$, some $\epsilon > 0$.

$$\sqrt{n} = O(n^{1/2 - \epsilon}), \text{ some } \underline{\epsilon > 0}$$

Fails

Case 2:- Is $\sqrt{n} = O(n^{1/2} * (\log n)^k)$,

a) Some $k \geq 0$?

$$\underline{\underline{k=0}} \quad \underline{\underline{\sqrt{n} = O(\sqrt{n})}}$$

Satisfied.

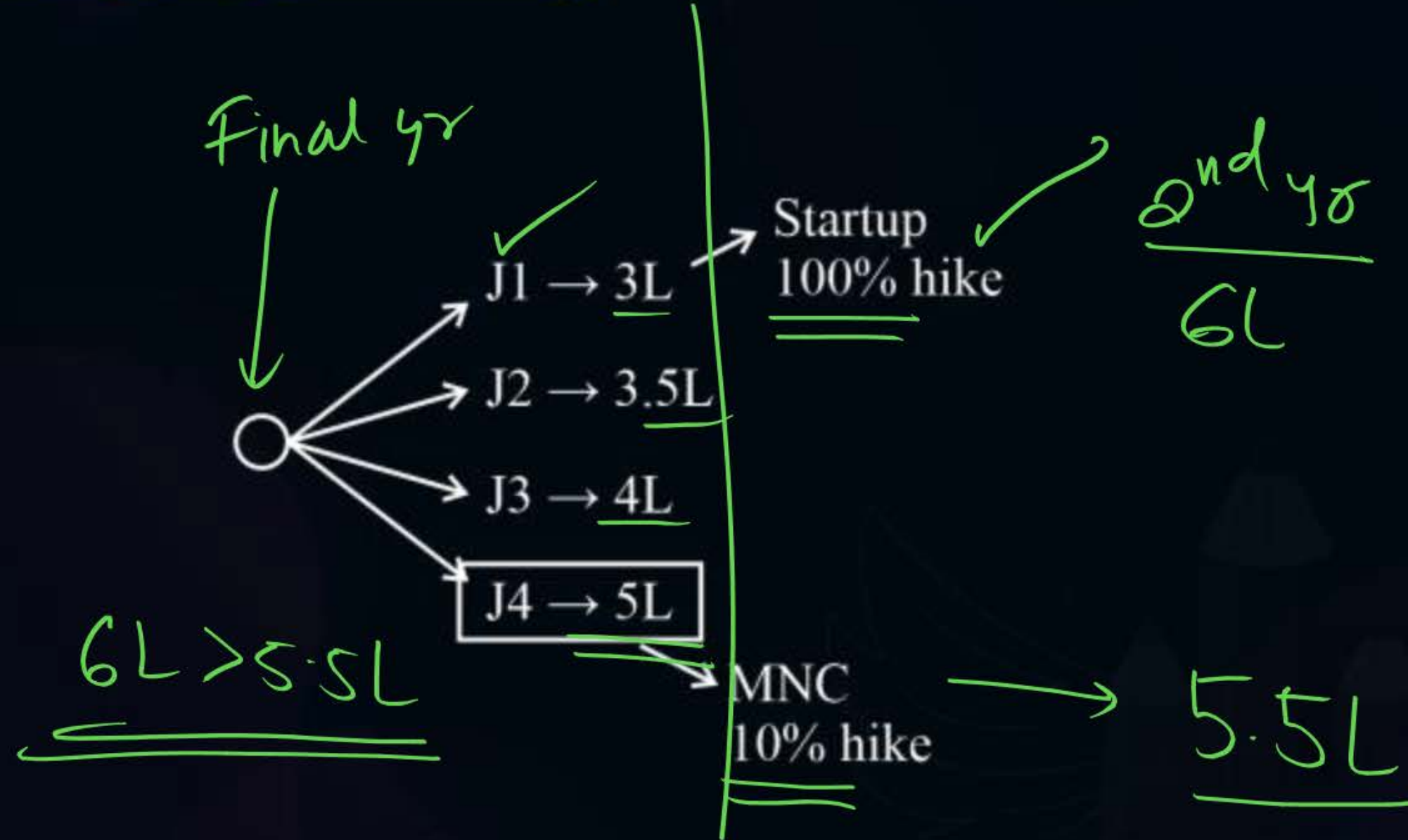
$$T(n) = O(n^{1/2} * (\log n)^{0+1}) = O(\sqrt{n} * \log n)$$

Greedy Algo



Topic : Greedy Algorithm

Idea:



Greedy



Knapsack

OMP

Huffman encoding

MST

PijKst to SSSP



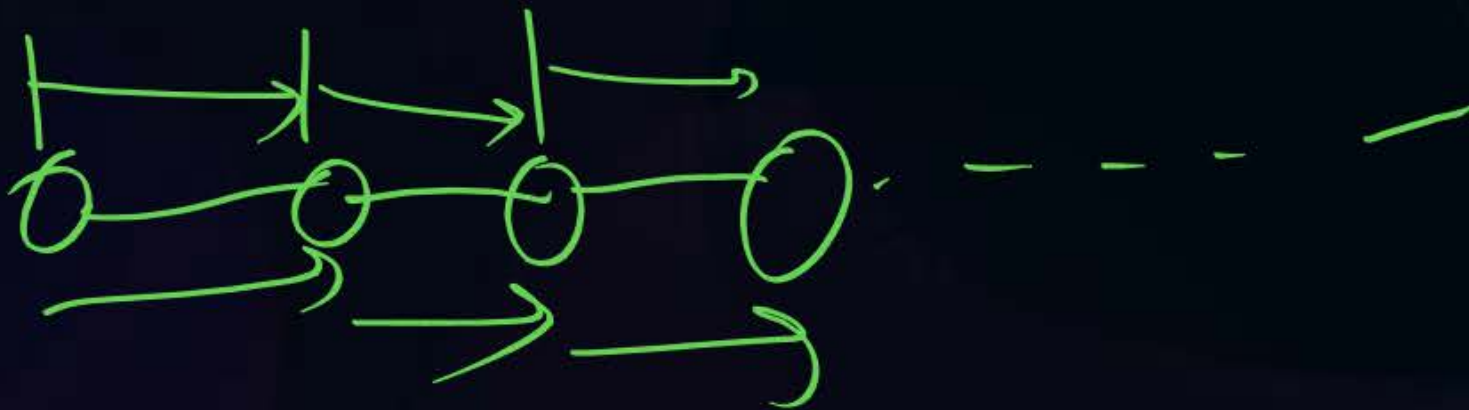


Topic : Greedy Algorithm



Steps:

- (1) Used for problems whose solutions are viewed as a set/sequence of decisions.
- (2) These decisions are made in a step-wise manner.
- (3) At each step, out of all the available options, gradually select ^{those} these option, which satisfies the criteria of the problem.



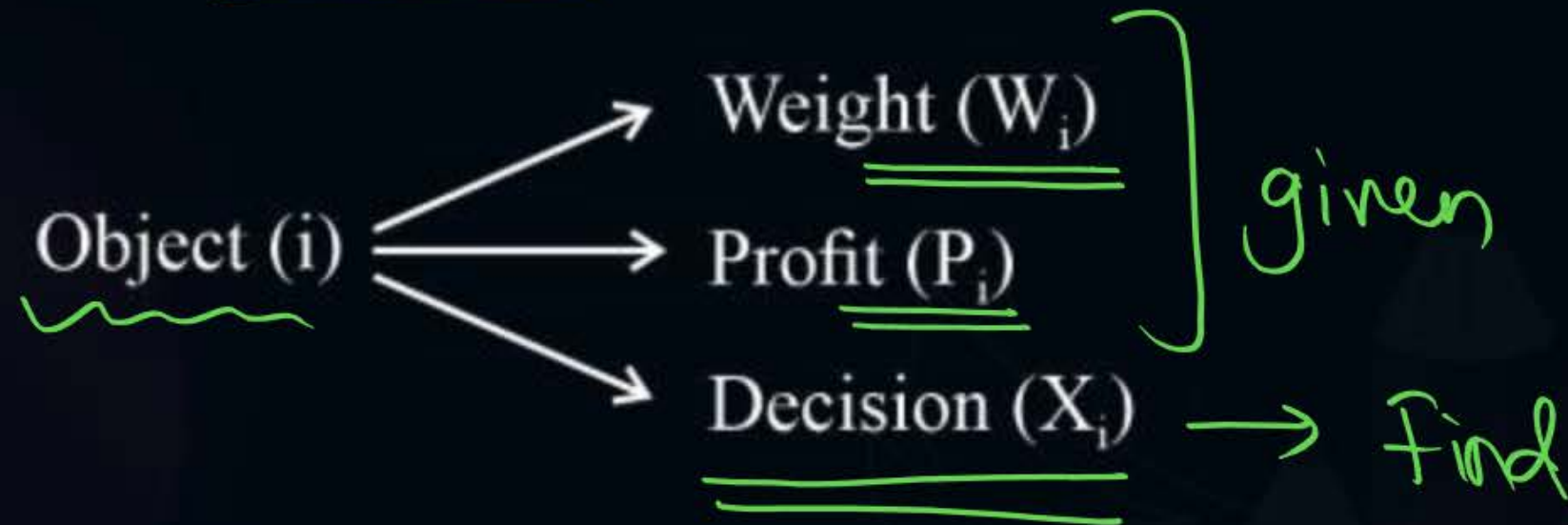
1) Knapsack Algo



Topic : Knapsack Problem

Description:

- (1) Given a Knapsack bag of capacity 'm'.
- (2) Given n objects:





Topic : Knapsack Problem

- You have to maximize the profit subject to the criteria that the total weight put into the knapsack does not exceed its capacity (M).

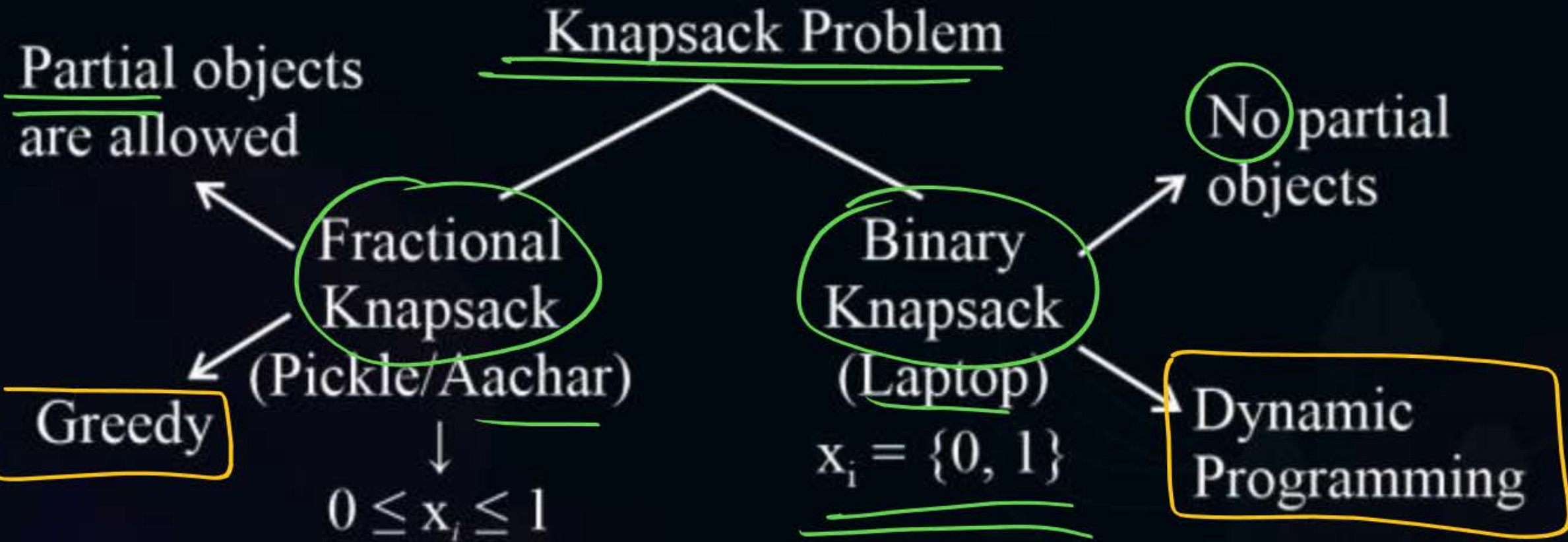
Criteria



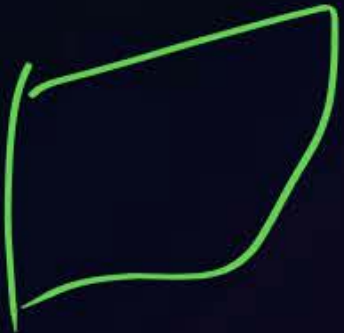


Topic : Knapsack Problem

V. imp



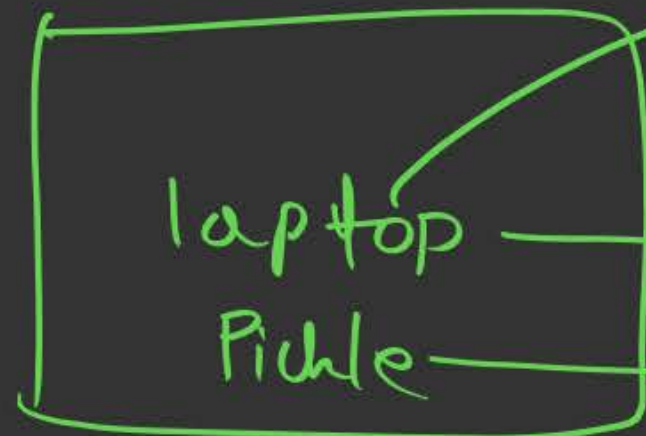
Us



5 kg



50% = 0.5



→ 3 kg

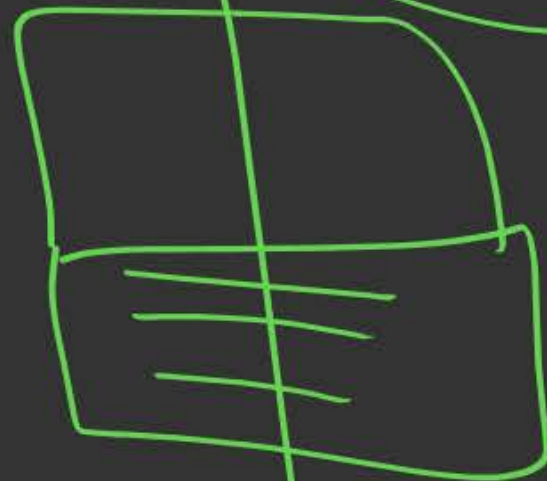
→ 5 kg

0/1 Knapsack

$X_i = \{0, 1\}$

2.5 kg overwt

$0 \leq X_i \leq 1$





Topic : Knapsack Problem

$O_1 \rightarrow P_1$	W_1	X_1
$O_2 \rightarrow P_2$	W_2	X_2
$O_3 \rightarrow P_3$	W_3	X_3

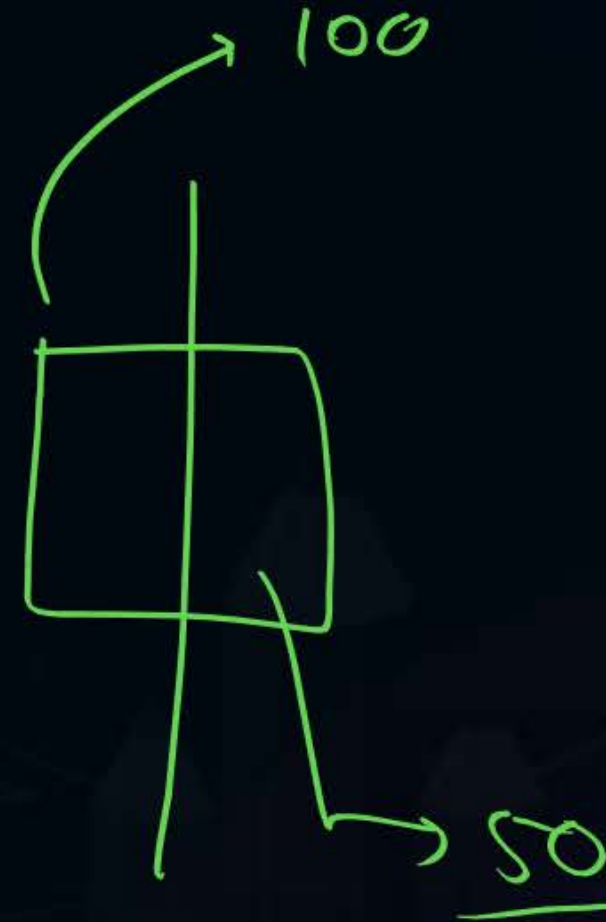
$\Sigma P_i \rightarrow \text{maximize}$ ✗

$\Sigma x_i * P_i \rightarrow \text{Maximize}$

Subject to criteria:

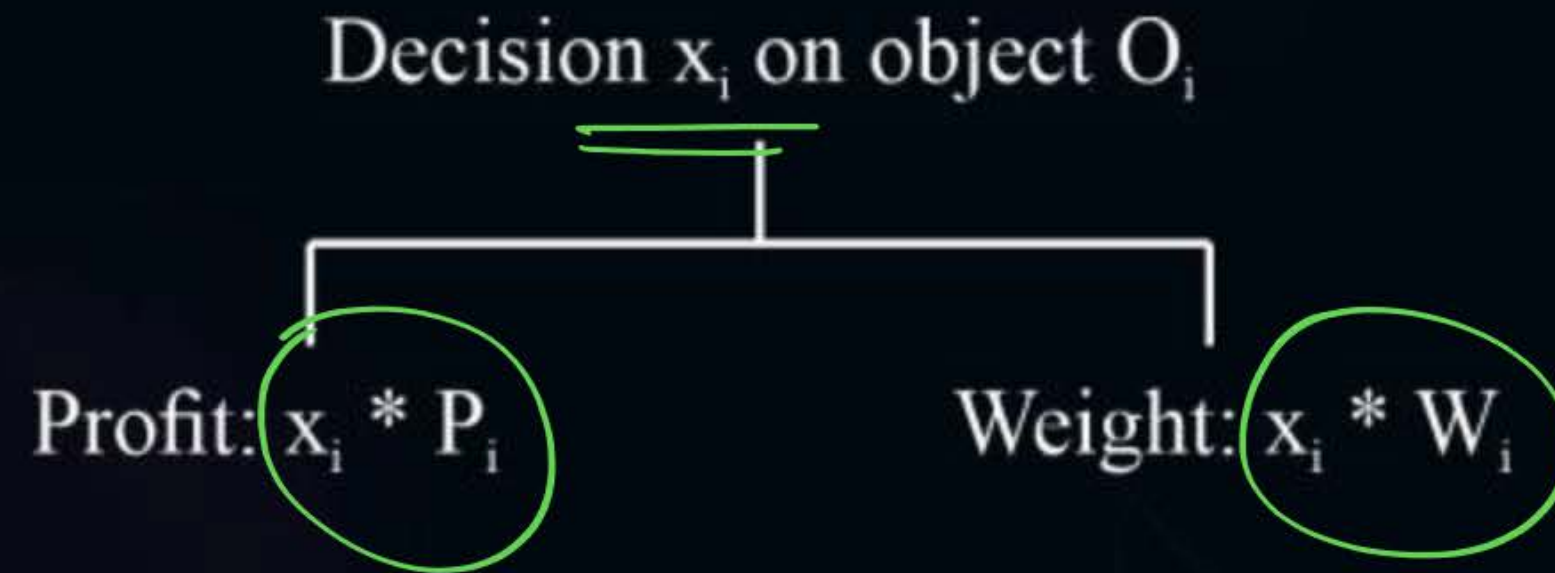
$$\Sigma x_i * W_i \leq M$$

$$\begin{aligned} X_i &= 1 \\ X_i &= 0 \\ X_i &= 50\% = 0.5 \end{aligned}$$





Topic : Knapsack Problem





Topic : Knapsack Problem

Example:

Fractional Knapsack:

No. of objects, $n = 3$

Knapsack Capacity, $M = 20$

$$[w_1 \ w_2 \ w_3] = [18, 15, 10]$$

$$[p_1 \ p_2 \ p_3] = [25, 24, 15]$$

01



- 1) Greedy about profit = 28.2
- 2) Greedy about weight
= 31
- ~~3) $P_i/w_i = 31.5$~~



Topic : Knapsack Problem

Aim: Maximize Profit:

Prefer objects with max Profit:

P_i : 25 24 15

W_i : 18 15 10

X_i : 1 $\frac{2}{15}$ 0

$$m = 20$$

$$m = 20 - 18 = \underline{\underline{2}}$$

15

$$1 \rightarrow \frac{1}{15}$$

$$x_2 = \frac{2}{15}$$

$$m = 0$$



Topic : Knapsack Problem

$$M = 20$$

$$18 < 20$$

$$M = 20 - 18 = 2$$

$$M = 2 - 2 = 0$$

Hence Total profit by this approach

$$= \sum_{i=1}^3 P_i * x_i$$

$$= P_1 \times x_1 + P_2 \times x_2 + P_3 \times x_3$$

$$= 28.2$$

$$m > w_i, x_i = 1$$



Topic : Knapsack Problem

Approach 2:

Greedy about weight:

P_i : 25 24 15

W_i : 18 15 10 ✓

X_i : 0 $\frac{10}{15}$ 1

$$20 > 10$$

$$m = 20 - 10 = 10$$

$$\frac{10}{15} = \frac{2}{3}$$

$$m = 10 - 10 = 0$$



Topic : Knapsack Problem

$$M = 20$$

$$M = 20 - 10 = 10$$

$$M = 10 - 10 = 0$$

Hence Total profit by this approach

$$= \sum_{i=1}^3 P_i * x_i$$

$$= P_1 \times x_1 + P_2 \times x_2 + P_3 \times x_3$$

$$= 31$$



Topic : Knapsack Problem

Approach 3:

Combines both logic (optimal):

01 $\rightarrow$ 1 P_1

02 $\rightarrow$ 1 P_2

03 $\rightarrow$ 1 P_3

1 $\rightarrow$ 2
1 $\rightarrow$ 5
1 $\rightarrow$ 3

Profit per unit wt

✓ $\frac{P_i}{W_i}$



Topic : Knapsack Problem

Weight adjusted Profits:

Profit per unit weight: P_i/W_i

$$W_1 \rightarrow P_1$$

$$1 \rightarrow P_1/W_1$$

$$W_2 \rightarrow P_2$$

$$1 \rightarrow P_2/W_2$$

$$W_3 \rightarrow P_3$$

$$1 \rightarrow P_3/W_3$$

$$X_1=0$$

$$P_1/w_1 = \frac{25}{18} = 1.4 \quad \text{--- (3)}$$

$$X_2=1$$

$$P_2/w_2 = \frac{24}{15} = 1.6 \quad \text{--- (1)}$$

$$X_3 = \frac{5}{10} = \frac{1}{2}$$

$$P_3/w_3 = \frac{15}{10} = \frac{3}{2} = 1.5 \quad \text{--- (2)}$$

Sort by P/W in desc order

$$20 > 15$$

$$M = 20 - 15 = 5$$

$$= 5 - 5 = 0$$



Topic : Knapsack Problem

$$M = 20$$

$$M = 20 - 10 = 10$$

$$M = 10 - 10 = 0$$

Hence Total profit by this approach

$$= \sum_{i=1}^3 P_i \times x_i$$

$$= P_1 \times x_1 + P_2 \times x_2 + P_3 \times x_3$$

$$= 0 \times 25 + 1 \times 24 + 0.5 \times 15 = 0 + 24 + 7.5 = 31.5$$

[NAT]

#Q. Find max/optimal solution to the fractional knapsack problem:

Given: $n = 7, M = 15$

$$15 - 4 = 11$$

$$11 - 5 = 6$$

$$6 - 2 = 4$$

① Greedy about Profit
 $\left[\begin{array}{l} \text{ } \\ \text{ } \end{array} \right]$

$$= 18 \times 1 + \frac{4}{7} \times 7$$

$$+ 1 \times 15 + 1 \times 10$$

$$= 18 + 4 + 10 + 15 = 47$$

Object	Weight	Profit
01	4	① 18 - 1
02	1	0 3 - 7
03	3	0 5 - 6
04	7	$\frac{4}{7}$ 7 - 4
05	1	0 6 - 5
06	5	① 15 - 2
07	2	① 10 - 3

[NAT]

#Q. Find max/optimal solution to the fractional knapsack problem:

Given: $n = 7, M = 15$

Object	Weight	x_i Profit
01	4 - (5)	1 18
02	1 - (1)	1 3
03	3 - (4)	1 5
04	7 - (7)	0 7
05	1 - (2)	1 6
06	5 - (6)	4/5 15
07	2 - (2)	1 10

① Greedy about weight

$$\begin{aligned}
 &= 18 + 3 + 5 + 6 \\
 &\quad + \frac{4}{5} \times 15 + 10 \\
 &= 42 + 12 \\
 &= 54
 \end{aligned}$$

$$\begin{aligned}
 15 - 1 &= 14 \\
 14 - 1 &= 13 \\
 13 - 2 &= 11 \\
 11 - 3 &= 8 \\
 8 - 4 &= 4
 \end{aligned}$$

[NAT]

#Q. Find max/optimal solution to the fractional knapsack problem:

Given: $n = 7, M = 15$

3) Optimal Soln?

P_i/w_i

Object	Weight	Profit
01	4	18
02	1	3
03	3	5
04	7	7
05	1	6
06	5	15
07	2	10



2 mins Summary



Topic

Topic

Topic

Topic

Topic

Ques on Masters mtd
Intro to Greedy Algo
Fractional Knapsack
 ↳ 1) Profit
 ↳ 2) weight
 ↳ 3) P/w.



THANK - YOU